Impact of Screen Time

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GitHub Repository Link: https://github.com/dg-vverma19/screen-time-impact-analysis-

Research Questions:

1.) How does productivity level vary by screen time bracket?
2.) How does attention span level vary by screen time bracket?
3.) How does screen time differ by education levels?
4.) Are certain age groups more likely to have a higher screen time?

Motivation:

I care about this specific issue, as it is an event that has rapidly increased in significance recently. Specifically, as society continues to technologically develop, and develop access to more online tools, screen time inevitably increases, whether it's for work, studying, or personal entertainment. However, screen time is also a major issue in the younger generation of this era, as I have personally noticed this. These research questions uncover where screen time is most common statistically, and its clear impact in multiple aspects. Knowing the answers to these affects our understanding of this development, by observing the possible positive and negative consequences, and predicting what future generations might be doing cognitively. Analyzing this can also facilitate research on if future devices can be modified to incorporate more positive features, rather than negative.

Dataset:

The dataset that I have chosen is from Kaggle and can be accessed through this link: https://www.kaggle.com/datasets/muhammadalirazazaidi/screen-time-data-productivity-and-attention-span?resource=download

Note: To download the dataset as a CSV file, one will have to sign into Kaggle first.

This dataset contains 15 descriptive columns including: age group, gender, education level, occupation, average daily screen time, device, screen activity, app category, screen time period, preferred work environment, productivity, attention span, work strategy, notification handling, and usage of products. For my research questions, I will not be utilizing all of these columns.

Overall, this dataset contains much data providing insight into the preferences and impact based on screen time, as well as the details about each respondent. This will overall allow for a deep analysis that will be effectively able to answer my research questions, as well as fulfill my motivation.

EDA Results:

Dataset Size/Missing Values:

Rows (199): Participants of Survey
Columns (16): Information about participants + details referring to screen time usage

This dataset does not contain any missing data, or NaN values. I determined this answer by creating a python module, containing a method that checks whether the given dataset file contains any NaN values. Specifically, I took use of a Pandas method: **file.isna().any().any().**

This call uses the first method, isna(), to mark each cell in the dataset a True value if it does not contain a value. Then, the first .any(), returns a Series contains True or False values representing if each specific column contains missing values or not. Finally, the second .any() creates a single value, checking overall if the full dataset contains any missing values or not.

False -> No missing values

True -> Missing Value(s)

In the case of my dataset, I ran this method in the main method, and found that this dataset does not have any missing values.

Dataset Variables/Columns of Interests:

Age Group:

> This represents the specific age of each participant in the dataset. I am using this specific column, as it ties directly into one of my research questions, that questions if certain age groups are to have a higher average screen time.

Average Daily Screen Time:

> This represents the average daily screen time that a given participant has in the dataset. I am using this specific column as it directly ties all of my research questions, as this is the essence of this entire project. This column ties into every research question, whether it's analyzing how this connects to age groups, productivity, etc.

Education Level:

> This represents the education level of a participant in the dataset, ranging from "High School or Below" to "Graduate." This ties into one of my research questions that asks how screen time differs based on educational level.

Attention Span:

This represents the attention span in different intervals that refers to one of my research questions investigating how screen time affects attention span.

Productivity:

This represents the productivity level in different intervals of participants that refers to one of my research questions that investigates how screen time intervals can affect productivity in participants.

*The summary of this dataset is provided in the *eda.py* module*

Challenge Goals:

Selected 2 Goals:

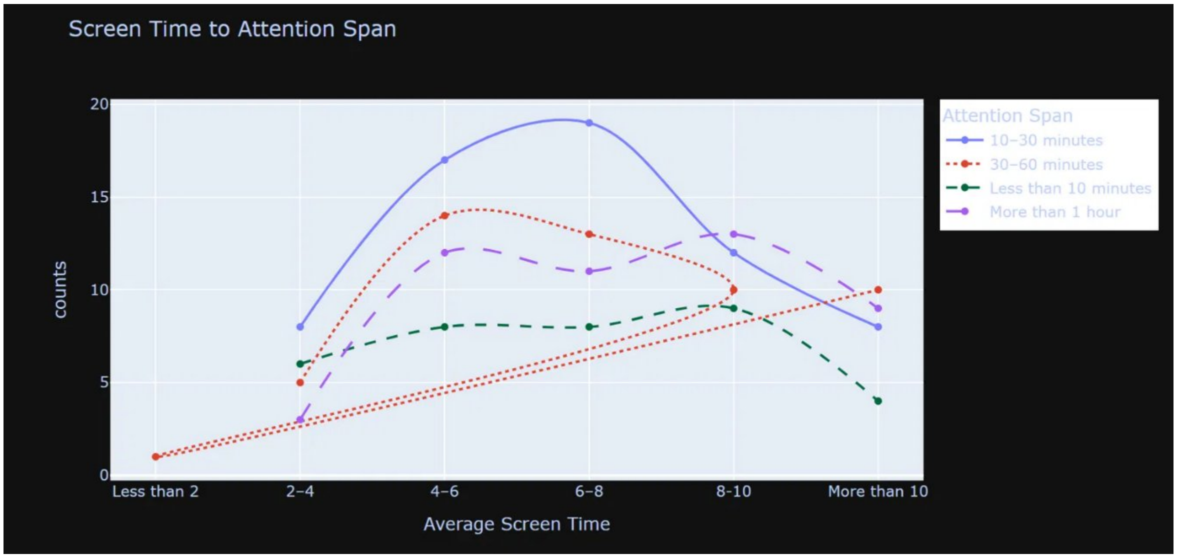
1.) Incorporating a new Python Library

I believe that this project will meet this goal the research questions developed for this project will require data visualization for effective analysis. By importing new types of data visualization libraries, this project will be to use more unique types of data visualizations to further enhance the effectiveness of the message conveyed. This project will meet this goal because it will further illustrate specific patterns about how screen time affects a variety of factors.

2.) Using Machine Learning

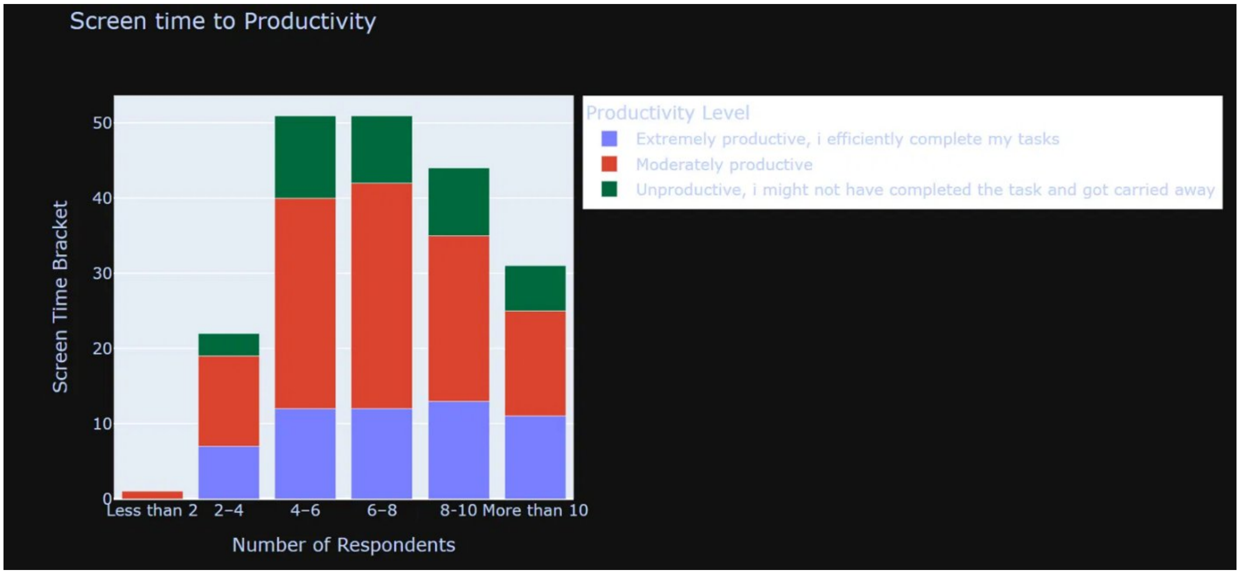
Part of this project is to analyze the specific decisions or preferences made based on both the details of a respondent, but also their screen time levels. With these variables, incorporating machine learning to predict possible scenarios using models will be effective on testing on how reliable, or how strong a trend is in this dataset. This project will meet this goal because it will further emphasize a specific trend, or pattern seen in this data.

Visualizations:



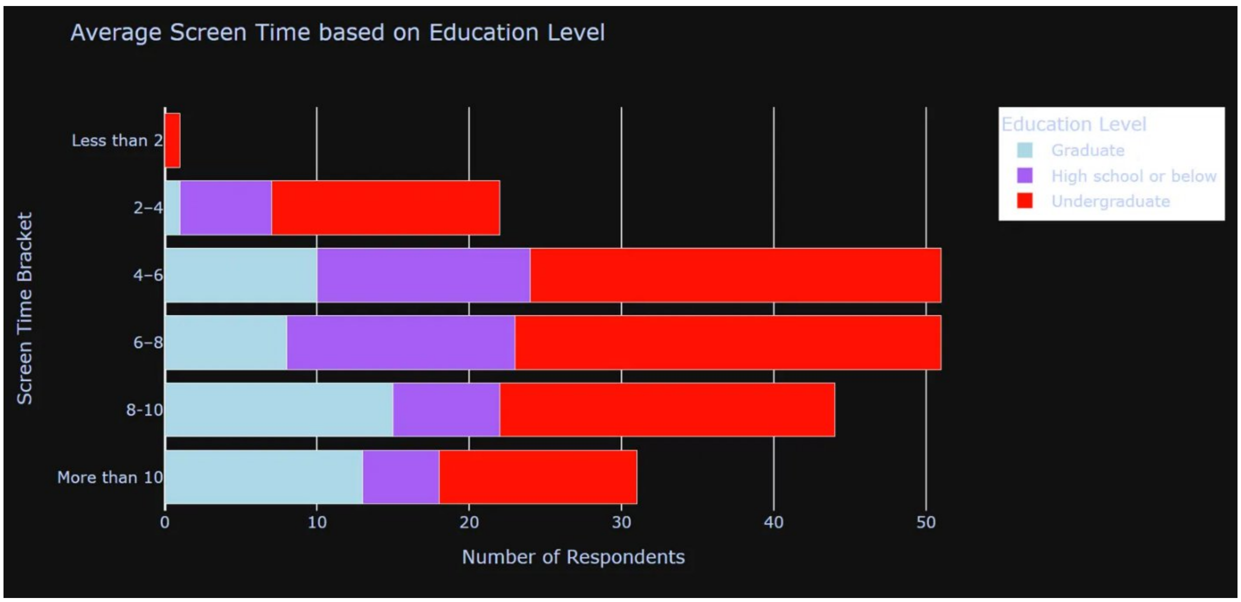
Caption: This graph is created to allow readers to interpret how average screen time fluctuates with attention span, and its main connection.

This line plot uses the variables of Average Daily Screen Time and Attention Span. These variables are based on a time interval that participants chose for both screen time, and attention span. A line chart was selected specifically to highlight trends across the ordered screen time brackets, which a bar or pie chart would not visually emphasize as clearly. One thing that readers should take away from this graph is how the 10–30-minute line decreases from the 6-8 to the 8-10 interval. This suggests a non-linear relationship for greater screen time hours compared to shorter attention spans. This also suggests that multiple factors might be at play for attention span, and not mainly just screen time.



Caption: This is a stacked bar graph that is meant to show how screen time brackets related to productivity levels.

This bar graph uses the variables Average Daily Screen Time and Productivity Level. These variables from the dataset are based on what participants chose from multiple intervals, whether it was a set of hours of hours for screen time, or a description for productivity. I chose this plot because it effectively shows the three productivity levels distributed on different screen time levels and emphasizes the counts. This graph does not show a clear pattern but indicates how the majority of the participants stayed moderately productive despite the number of screen time that they had. Despite that, the most productive participants stayed in the 4-6 hour screen time interval, which something key that readers should take away.



Caption: This horizontal stacked bar graph highlights how education level affects the daily average screen time.

This bar graph uses the variables of Education Level, and Screen Time Bracket. Education level ranges from high school or below, to graduates. Screen Time Brackets range all the way to 2 hours, to more than 10 hours. This bar graph was chosen because it can visually be seen the number of respondents in each education level, and their average screen time. The main thing readers should take away from this graph is that that the majority of the data is undergraduate, and they appear the largest in most of this graph. However, the blue section, or graduates, start increasing as screen time increases, opposite of the pattern with undergraduates. This hints at a pattern or relationship between education level and average screen time, which readers should take away.

Overall, these visualizations significantly contribute to my EDA. These visualizations provide key insight into the relationships between data, and how there might not be extremely clear patterns between the variables that I decided to investigate. These also showed me what the majority of the data in the dataset is, and showed me some key distinct patterns, that I can use for further analysis. These visualizations were all produced using the python library plotly, and making use of three of its visualizations: vertical stacked graphs, line plot, horizontal stacked graphs. Then, after creating each visualization, and creating customizations, each graph was saved as an HTML file.

Testing:

To verify accuracy, I compared the total count from each grouped table used in my visualizations against the total number of rows in the original dataset, confirming they matched for all three groupings. The logic is that since I would group my dataset by two columns for each of my visualizations, I would simply re-organize my dataset. But essentially, if I use .size() and .sum() to check how many rows are in each group, that should equal back to the original amount of rows in the dataset. While this doesn't check exact values, it checks if I made any filtering errors, or if I accidently removed something from the data. Despite that, I never changed the value of anything, so it wouldn't make sense to check the exact values of each visualization, checking if it exists in the data.

Work Plan Evaluation:

My work plan worked exceptionally well, but it took a little less time than I had originally planned. Plus, due to some issues during code, I was forced to change some of the methodology and visualizations that I was creating. For example, instead of creating multiple pie charts, that would be extremely inconvenient, I created a simple bar graph that evaluated everything at once that was much more efficient. This did take a little more research and planning, however once this one step was implemented, the rest of the steps were much easier and faster to do, making my each step estimated time an overestimate. However, this plan did help my complete ¾ steps in my project and leaving me with my final step. This step will challenge me to use extended machine learning concepts, that I believe will take the same amount of time that I stated in my proposal.